

TITELBILD

RUHR-UNIVERSITÄT BOCHUM

PROGRESS MEETING

Exploration vs Exploitation

MiniWorld reproduction + MiniGrid extension

Challenging Problems in Reinforcement Learning · SS 2026

Progress meeting goal

Discussion of results, report, and final presentation

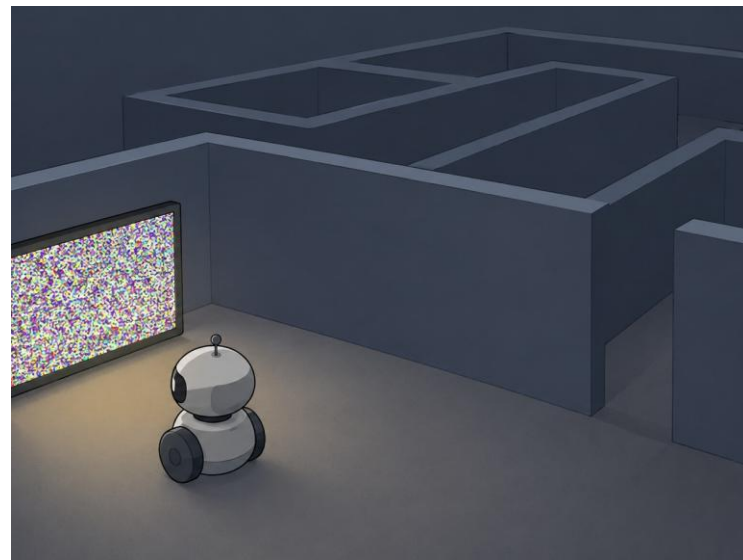
- MiniWorld reproduction is complete: paper setup, 64 seeds, noisy-TV analysis.
- MiniGrid extension is complete: PPO with none / entropy / RND / LPM.
- Main story: coverage is not enough; task success and robustness are more informative.
- Today: agree on interpretation, caveats, and final presentation emphasis.

Status: experiments completed; focus now shifts to narrative, caveats, and slide selection.

MiniWorld reproduction: setup

Goal: reproduce the LPM paper's noisy-TV robustness claim

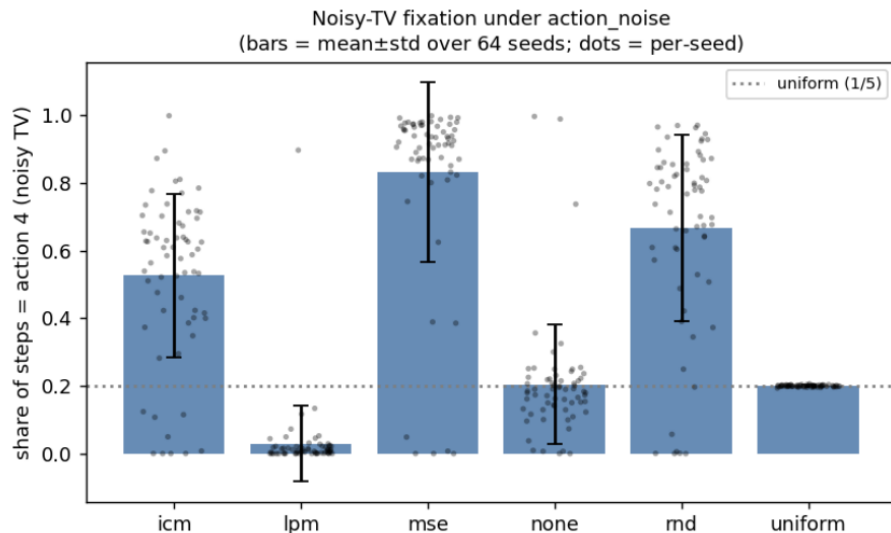
- 3D MiniWorld maze with no extrinsic reward.
- Variants: nonoise, noisy_tv, action_noise.
- Methods: LPM, RND, ICM, MSE, none, uniform-random.
- A2C, shared CNN encoder, 50k steps, 64 seeds.
- Metrics: coverage, TV-action share, coverage heatmaps.



This block validates the noisy-TV mechanism before moving to sparse rewards.

MiniWorld: noisy-TV robustness

TV-action share under action_noise measures direct fixation on the distractor

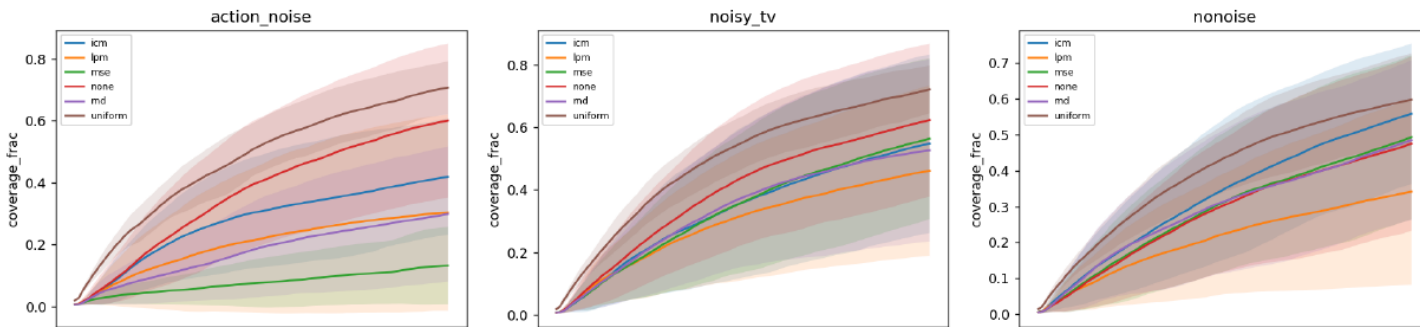


- MSE fixates strongest: about 83% of steps.
- LPM almost never selects the noisy action: about 3%.
- RND and ICM are partially fooled.
- Central noisy-TV robustness claim reproduces.

LPM rewards improvement, not surprise itself.

MiniWorld: coverage metric issue

Coverage curves do not show the same ranking as the paper



- Uniform-random covers the most area in all variants.
- LPM avoids the noisy wall, but does not obtain the highest coverage.
- Coverage rewards undirected movement in this small extrinsic-free maze.
- Noise robustness and useful exploration are decoupled.

Bridge: what is “good exploration”?

Reason for the second experiment block

- MiniWorld has no task reward, so coverage is only an indirect proxy.
- A high-coverage policy can still be random and not useful.
- Sparse-reward MiniGrid asks: does exploration help solve the task?
- Therefore, the extension uses final evaluation return / task success.

Project pivot: from “visited cells” to “solved sparse-reward tasks”.

MiniGrid sparse-reward study: setup

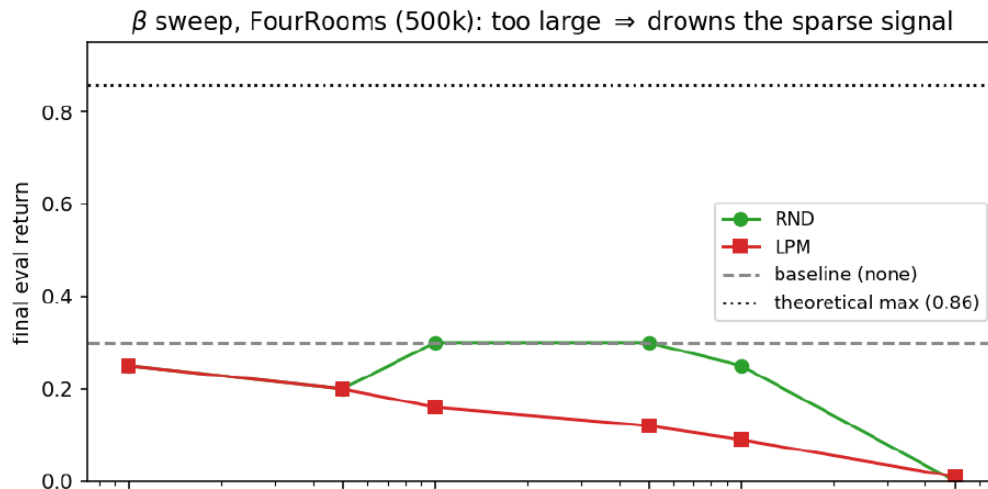
PPO on symbolic 7×7 observations; clean and noisy layout distributions

- Difficulty ladder: DoorKey-5x5, FourRooms, MultiRoom-N6.
- Base learner: PPO MlpPolicy on flattened 7×7×3 symbolic image observation.
- Methods: none, entropy, RND, LPM.
- Noise: global cell-level corruption of the egocentric symbolic view.
- Evaluation: deterministic return on randomized layouts; no intrinsic bonus in eval.

Caveat: MiniGrid noise is global observation noise, not a localized noisy-TV wall.

MiniGrid: β sweep

Intrinsic reward coefficient β must be tuned

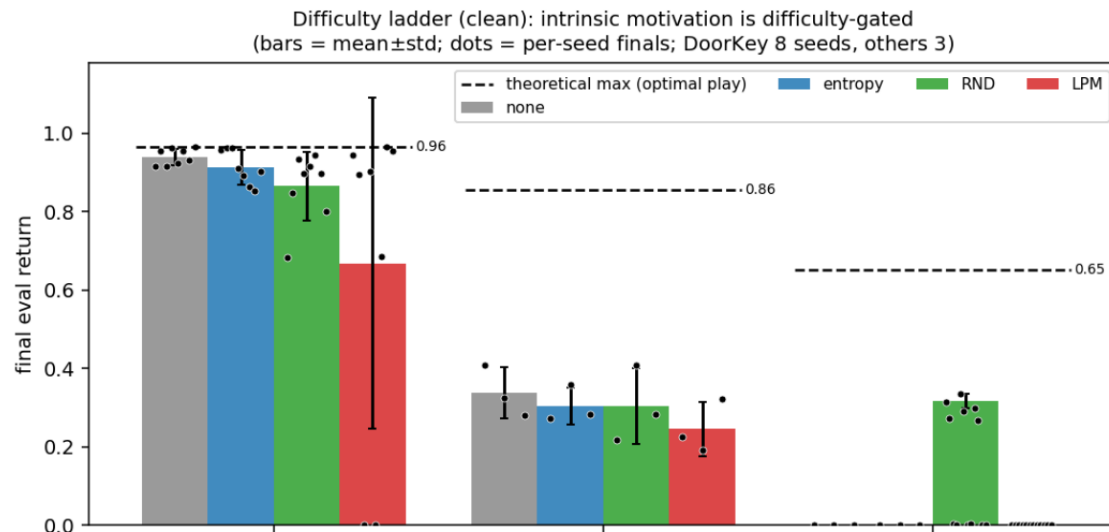


- β too small: intrinsic reward is mostly ignored.
- β too large: intrinsic reward drowns sparse task reward.
- Usable range is roughly 0.001–0.005.
- At $\beta = 0.05$, performance collapses.

Intrinsic motivation is not plug-and-play.

MiniGrid: clean difficulty ladder

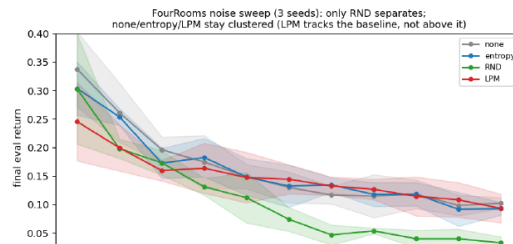
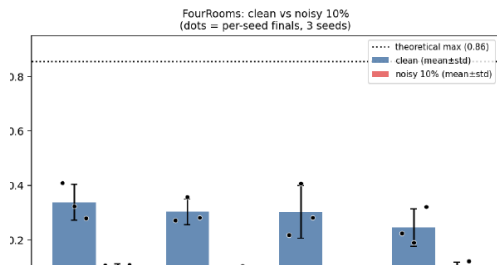
Where does intrinsic motivation actually help?



- Easy DoorKey: baseline already solves.
- Medium FourRooms: all methods remain far from the ceiling.
- Hard MultiRoom-N6: only RND reaches the goal.
- Clean setting: RND explores more aggressively than LPM.

MiniGrid: noise robustness

Strong exploration can become fragile under observation noise



- DoorKey 10% noise: LPM solves all 8 seeds; RND becomes highly variable.
- FourRooms sweep: RND degrades fastest and separates from the pack.
- None, entropy, and LPM remain statistically close.
- LPM avoids RND's noise-chasing failure, but does not clearly beat the plain baseline.

Trade-off: RND = exploration power; LPM = noise robustness.

Discussion points for today

Align the report and final presentation narrative

- Narrative: MiniWorld explains the metric problem; MiniGrid gives the task-success trade-off.
- Main claim: LPM is not “better exploration” in general; it is better framed as noise-robust intrinsic reward.
- Caveat: MiniGrid noise is global observation corruption, not the exact localized noisy-TV setup.
- Caveat: some results use only 3 seeds; DoorKey was increased to 8 seeds for the robustness check.
- Final presentation: MiniWorld as motivation, MiniGrid as main contribution.

RND buys hard-exploration ability at the price of noise-fragility; LPM trades this aggressiveness for robustness.